Unit wise Question set DBMS (CACS255)

(Past Questions)

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Unit 1: Introduction to DBMS

1. **[2020 Q2:]** What is DBMS? Describe the merits of using DBMS. **[5]**
2. **[2021 Q2:]** What is database? Discuss the importance of DBMS. **[5]**
3. **[2022 Q2:]** What is DBMS? Describe the advantages and disadvantages of using DBMS. **[5]**
4. **[2023 Q2:]** Define database and DBMS. Explain the benefits of DBMS in detail. **[5]**
5. **[2024 Q2:]** Define database system. Write the merits and demerits of DBMS. **[5]**
6. **[2025 Q2:]** What is flat file system? Explain the advantages of using DBMS over flat file system. **[5]**

Unit 2: Database Design, Architecture and Model

Three-Level Architecture & Data Independence

1. **[2022 Q3:]** Describe 3-schema architecture of database. **[5]**
2. **[2024 Q3:]** Draw and explain three schema architecture of the database system. **[5]**
3. **[2025 Q3:]** How logical data independence is different from physical data independence? Explain three level database architecture. **[5]**
4. **[2026 Q3:]** Explain physical data independence and logical data independence. **[2.5+2.5]**

Data Models

1. **[2020 Q3:]** What is Data Model? What is the difference between hierarchical and network data model? What are its types? **[5]**
2. **[2022 Q11:]** What is data model? What are its types? Describe the structure of database. **[10]**
3. **[2026 Q6:]** What is data model? What are its types? **[1+4]**

DDL, DML and Database Languages

1. [2026 Q9:] Describe DDL and DML queries with example. Explain database system architecture. [3+7]

ER Diagrams & Entity Sets

1. **[2022 Q7:]** Describe the use of ER diagram. Draw an ER diagram for a university to design a database for it. **[5]**

2. **[2024 Q4:]** What are an entity and its attributes? Write the difference between a strong and weak entity set with an example. **[5]**
3. **[2024 Q10:]** How participation constraints are used in ER diagram? Explain. Also construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period and has an associated due date, and the date when the payment was received. **[10]**
4. **[2021 Q10:]** a) Design an ER diagram of your choice. The ER diagram should contain at least five entities. Out of those entities, one should be a weak entity. Your entities should have appropriate attributes including primary key attributes, if any. There should be presence of one to one relationship, one to many relationship and many to many relationships in your ER diagram. **[10]**
   a. b) Convert the ER diagram in (a) into equivalent relational tables.
5. **[2020 Q9:]** Design an ER diagram for Hospital System. Use your assumption for the selection of attributes, entities and relationships. Show the use of total and partial participation along with the appropriate cardinalities. **[10]**
6. **[2025 Q10:]** How ER model can be mapped to the relational data model? Explain with a proper example. **[10]**

Unit 3: Relational Database Model

Keys and Integrity Constraints

1. **[2023 Q5:]** How candidate key is different from primary key? Also explain foreign key in detail. **[5]**
2. **[2024 Q5:]** Why integrity constraints are used in databases? Explain different types of integrity constraints with examples. **[5]**
3. **[2025 Q4:]** How primary key is different from super key? Explain composite key and partial key. **[5]**

Relational Algebra

1. **[2021 Q3:]** Define union compatibility. Given following relations, show the resulting relation of Director ∩ Actor and Actor ∪ Director. **[5]**

Director

Fname	Lname
Deepak	Giri
Bhuwan	KC
Dipendra	Khanal

Actor

First Name	Last Name
Deepak	Giri
Rajesh	Hamal
Dipendra	Khanal

2. **[2022 Q8:]** What is DDL and DML queries? Consider the following relation

Customer Name	Loan Number
X	L01
Y	L02

Relation borrower

Loan Number	Branch Name	Amount
L01	B1	5000
L02	B2	6000

Relation loan

Write a relational algebra expression to find all customers who taken loan from branch "B1". **[5]**

3. **[2023 Q4:]** What is relational algebra? Explain different RA operations. **[5]**
4. **[2025 Q5:]** What are selection and projection operators in relational algebra? Given a table with schema: **[5]**
    a. Student(Rollno(PK), Fname, Lname, Address, Level, Did(FK))
    b. Department(Did(PK), Dname, HOD)
    c. Write an RA query to find the name of students who are in 'Bachelor' level living in 'Birgunj' and studies under the department with department head 'Ramesh Sharma'.

Unit 4: Database Normalization

Functional Dependencies

1. **[2023 Q9:]** What are fully and partial functional dependencies? Explain 2NF and 3NF in detail. **[10]**
2. **[2024 Q6:]** What is functional dependency? Explain its types with examples. **[5]**

Normal Forms (1NF, 2NF, 3NF, BCNF)

1. **[2020 Q10:]** Discuss the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples. **[10]**
2. **[2021 Q5:]** Why normalization is needed? When any relation is said to be in 1NF and 2NF? **[5]**
3. **[2022 Q9:]** Define normalization. Describe the importance of normalization in DBMS. Describe 1NF, 2NF and 3NF with examples. **[10]**
4. **[2024 Q9:]** Why normalization is required in database? Explain the 1NF, 2NF and 3NF of normalization with example. **[10]**
5. **[2025 Q9:]** What are the advantages of normalization? Explain 3NF and BCNF with example. **[10]**
6. **[2026 Q10:]** What is normalization? What are its advantages? Describe its types with appropriate examples. **[2+2+6]**

Unit 5: Creating and Altering Database and Tables (SQL)

SQL Constraints and DDL

1. **[2023 Q3:]** What are different SQL constraints to specify the rules for the data in the database? Explain. **[5]**

Unit 6: Manipulating and Querying Data

SELECT, ORDER BY, GROUP BY, HAVING

1. **[2021 Q4:]** How ORDER BY and GROUP BY HAVING Clause in SQL are different? Illustrate with suitable examples. **[5]**

JOIN Operations

1. **[2023 Q10:]** How different join operations can be performed? Explain with proper examples. Consider a database of a shipment company. Shipped items are received into the system at a single retail center. Shipped items make their way to their destination via one or more standard transportation events (i.e. flights, truck deliveries). The transportation events follow a regular schedule. Create an ER diagram to represent this information. **[10]**
2. **[2025 Q11:]** a) What is inner join? Perform inner join to find the customer names who are going on flight to 'South Wales': **[10]**

    Customer(Cid, Name, Address, Age, Email)

    Booking(Cid(FK), FlightID(FK), BookedDate)

    Flight(FlightID, AirplaneCode, DepartureTime, Destination, Duration)

    b) What are unary and ternary relationships? Write with examples.

SQL Queries (Practical / Schema-based)

1. **[2020 Q11:]** Consider a database system with following schemes: **[10]**

    Restaurant(Rname, Rlocation, Rfund)

    Cook(Cname, Cspeciality)

    Worksat(Cname, Rname, Workinghrs, Shift)

    Food(Fname, Cname, Catagory)

    Write SQL statements and relational algebra statements for following queries:

    a) Select the name and location of all restaurants.

    b) Find the working hours of cook named "Sita".

    c) Select name of the foods cooked by "Ramesh".

    d) Use join to select name of restaurants where food of category "breakfast" is available.

    e) Find the name of cooks who work at "KFC".

2. [2021 Q9:] Given the schema as below, write relational algebra and SQL queries for following: **[10]**

Sailors(sid, sname, rating, age)

Boats(bid, bname, colour)

Reserves(sid, bid, reservedate)

a) Find name, rating and age of all the sailors.

b) Find the name and color of boats having color blue.

c) Find name of the boats reserved on date 2021/07/20.

d) Find the names of boats and sailors who have reserved boats having color green.

e) Use Left and Right Outer Join between Sailors and Reserves.

3. [2023 Q11:] Consider the following tables: **[10]**

Book(Bookid, Title, Publisher, PublishedDate, Price)

Book_Authors(Bookid, Author name)

Borrower(CardNo, Name, Address, Contact)

Bookissue(BookId, CardNo, IssuedDate, DueDate)

Write SQL query for the following:

a) Find the name of the books taken by Ramesh Sharma.

b) Find the issued book name and the borrower's name who have issued date before 15th May, 2023 and due date after 15th June, 2023.

c) Find the borrower's name who has taken the book written by Laxmi Prashad Devkota.

d) Find the number of books issued on the date 25th April, 2023.

4. **[2024 Q11:]** Consider the following employee database, where primary keys are underlined: **[10]**

Supplier(supplier-id, supplier-name, city)

Supplies(Supplier-id, parts-id, quantity)

Parts(part-id, par-name, color, weight)

a) Write a SQL query to create above tables.

b) Find the name of all suppliers who have supplied parts with quantity more than 100 and the name of all parts supplied by "RD Traders".

c) Find the number of parts supplied by "S02".

d) Find the name of suppliers located in city "KTM".

5. **[2026 Q11:]** Consider the following schema: **[3+2+2+1+2]**

Guest(Gid, Gname, Age, Gender)

STAFF(Sid, Sname, Age, Gender, Roles)

SERVICE(serIid, serName, Price)

ASSISTS(Gid, Sid) – Which staff assists which guest

PROVIDES(serIid, Sid) – Which staff provides which service

Write SQL Queries:

a) Find name of guests who are VIP.

b) Find total number of guests assisted by staff Ram.

c) Find name of staff whose name starts with A and ends with A.

d) Display name of staff who provides laundry service.

e) Give 10% discount to female guests on the price of service they receive.

Unit 7: Developing Stored Procedures, DML Triggers and Indexing

Stored Procedures

1. **[2020 Q8:]** What is the advantage and disadvantage of using stored procedure? How can you create and execute stored procedure? **[5]**
2. **[2023 Q6:]** What is a stored procedure? How can it be used in a database? Explain with an example. **[5]**
3. **[2025 Q6:]** How parameters can be passed to a stored procedure? Explain with an example. **[5]**
4. **[2026 Q4:]** What is stored procedure? Describe statement level trigger and multi row enabled trigger. **[1+4]**

Triggers

1. **[2021 Q11:]** Define stored procedures and triggers. How they are applicable in database? Illustrate with examples, how can you create before and after triggers on INSERT query. **[10]**

Indexing

1. **[2020 Q4:]** Why indexing is essential in database? Differentiate dense index from sparse index with suitable example. **[5]**
2. **[2022 Q5:]** Why indexing is essential in database? Differentiate dense index from sparse index with suitable example. **[5]**
3. **[2025 Q8:]** What are different types of indexing used in database? How index can be created using SQL? Write with example. **[5]**
4. **[2026 Q8:]** Why indexing is essential in database? Differentiate dense index from sparse index with suitable example**.** **[5]**

Unit 8: Query Processing and Security

Query Processing and Optimization

1. **[2020 Q5:]** Discuss about materialized and pipelined evaluation of query expression in query optimization. **[5]**
2. **[2021 Q6:]** How cost of query is measured during query processing? Construct query expression tree for the relational algebra query: **[5]**
    a. $\pi_{FName,LName,Address}(\sigma_{DName="Research" \wedge Department.Dnumber=Dno}$ (Employee ⋈ Works_on))
3. **[2023 Q8:]** What is database security? Describe steps of query processing. **[5]**
4. **[2026 Q7:]** Define query processing and steps in query processing. **[1+4]**

DBA Roles and Responsibilities

1. **[2022 Q4:]** What are roles and responsibilities of DBA? **[5]**

Database Security and Access Control

1. **[2024 Q7:]** Differentiate between discretionary access control and mandatory access control with example. **[5]**

Unit 9: Transaction and Concurrency Control

Transactions and ACID Properties

1. **[2021 Q8:]** How read and write operations are performed in a transaction? List the various states where transactions can enter into. **[5]**
2. **[2022 Q6:]** What are the states of transaction model? Describe the ACID properties of transaction. **[5]**

3. **[2024 Q8:]** Write the ACID properties of transactions. Explain the concepts of serializability in database transactions. **[5]**
4. **[2026 Q2:]** Define transaction. Describe ACID properties with example. **[1+4]**

Serializability and Schedules

1. **[2026 Q5:]** Define schedule and serializability with example. How can you test the serializability? **[2+3]**

Concurrency Control

1. **[2020 Q6:]** Explain with examples, how lost update and dirty read problems can occur in the context of transactions. **[5]**
2. **[2020 Q7:]** How timestamp ordering protocol is used to ensure concurrency control? **[5]**
3. **[2021 Q7:]** How wait-for-graph is constructed? How can you use it to detect deadlock in a schedule of transactions? Support your answer with an example. **[5]**
4. **[2022 Q10:]** Differentiate between stored procedure and trigger. Why concurrency control is needed? Describe timestamp based concurrency protocol. **[10]**
5. **[2023 Q7:]** What are the major purposes of concurrency control? Explain two phase locking protocol in detail. **[5]**
6. **[2025 Q7:]** How timestamp based algorithm uses timestamp to serialize the execution of concurrent transactions? Explain in detail. **[5]**

Note:

- The year mentioned in the questions is the year the exam was conducted.

Sources:

1. www.unibytes.xyz, https://unibytes.xyz/database-management-systemdbms-board-questions-collections/ www.tupapers.com, https://www.tupapers.com/course/bca/fourth-semester/database-management-system/
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